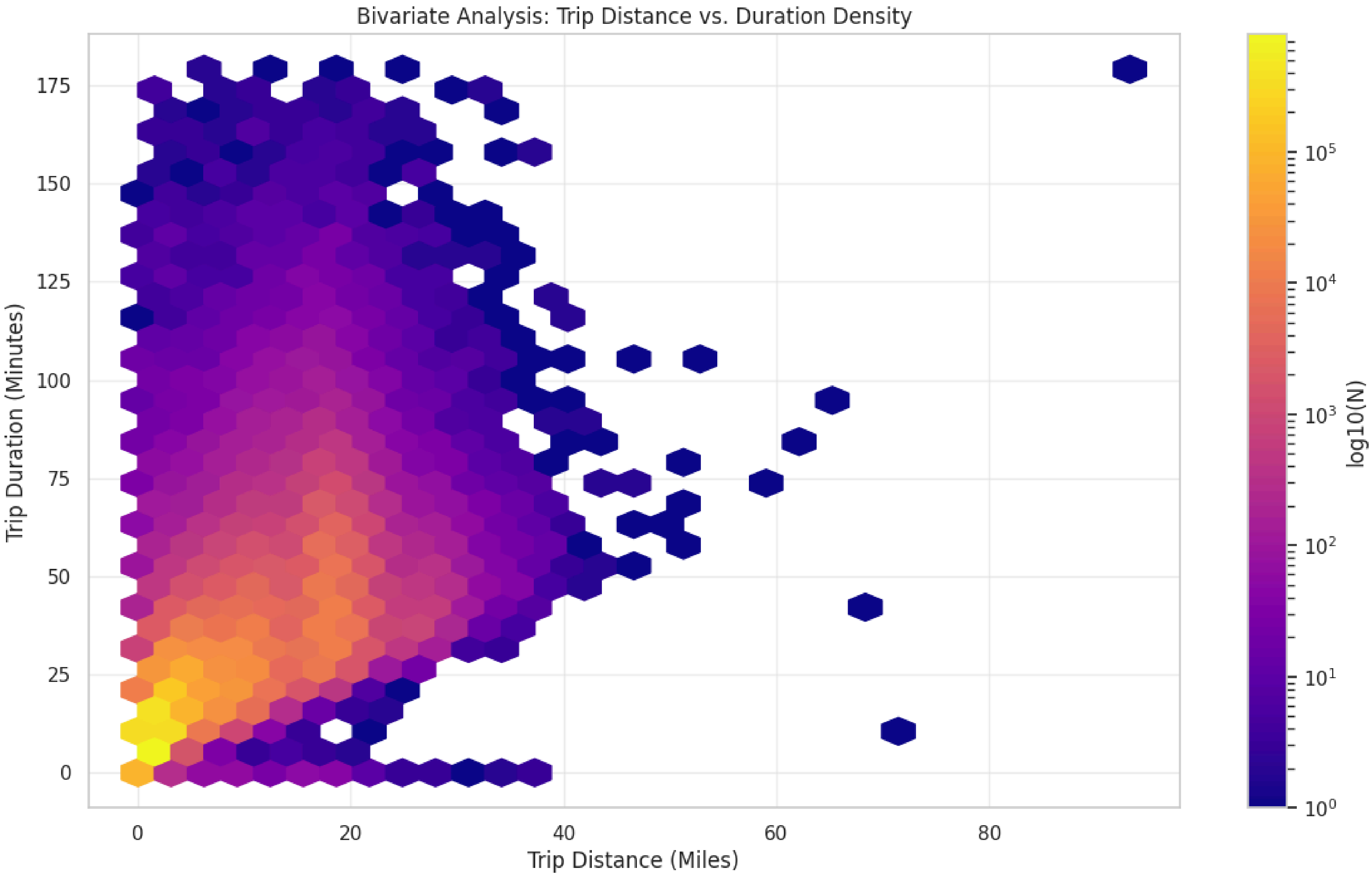
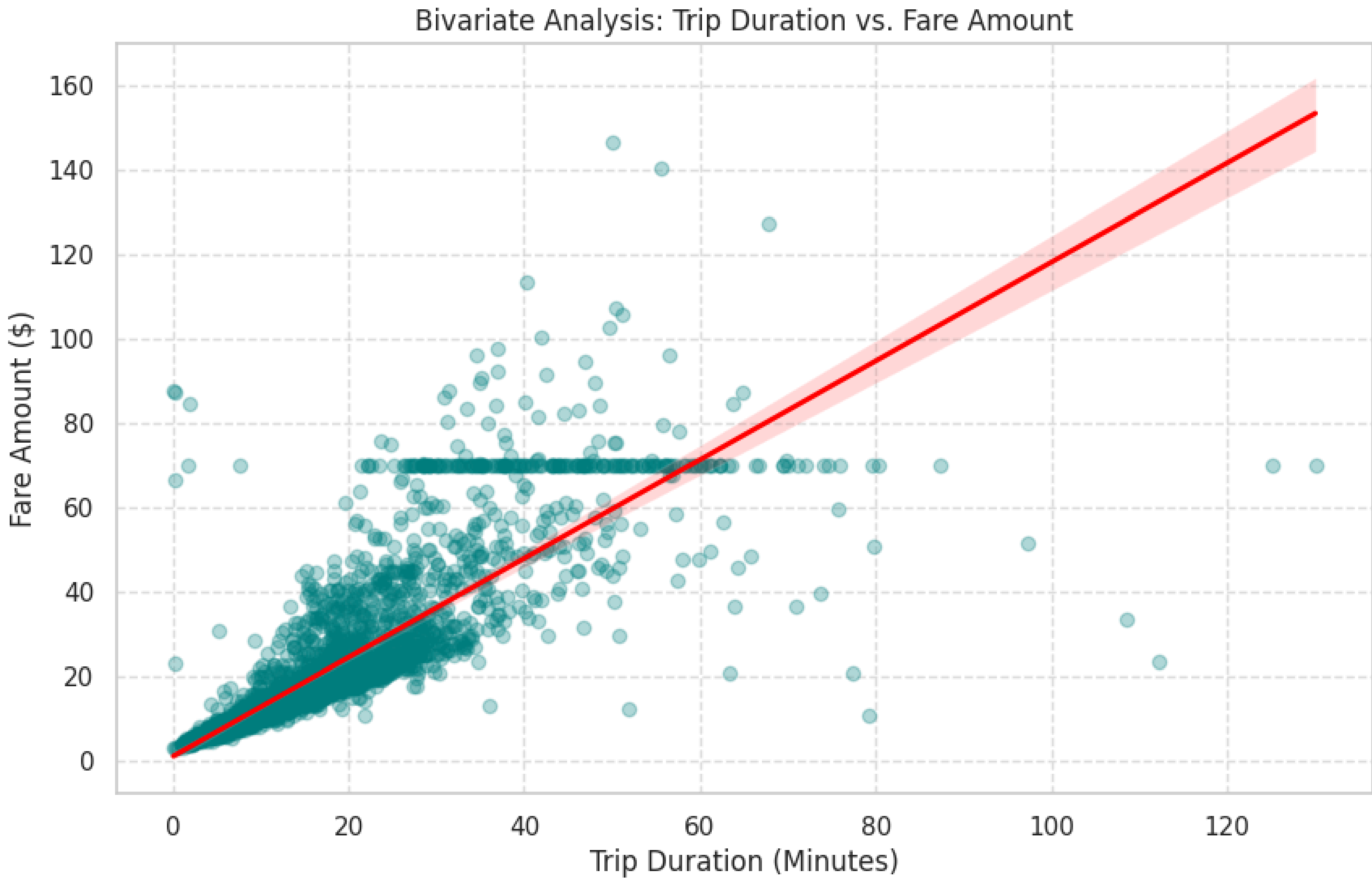
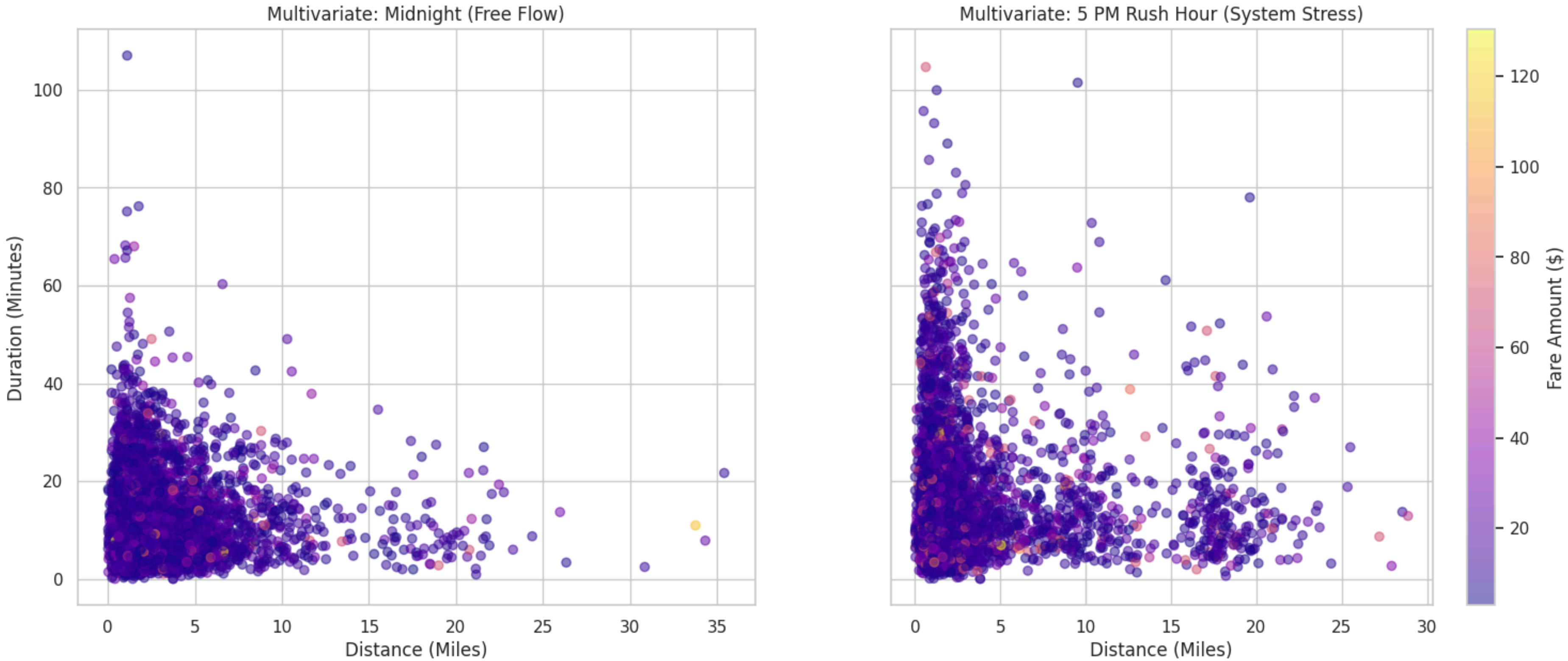
Smart Intelligent Traffic Management System: Uncovering existence of traffic congestion in NYC
Dataset source: "https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2024-01.parquet"
Dataset dictionary: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page



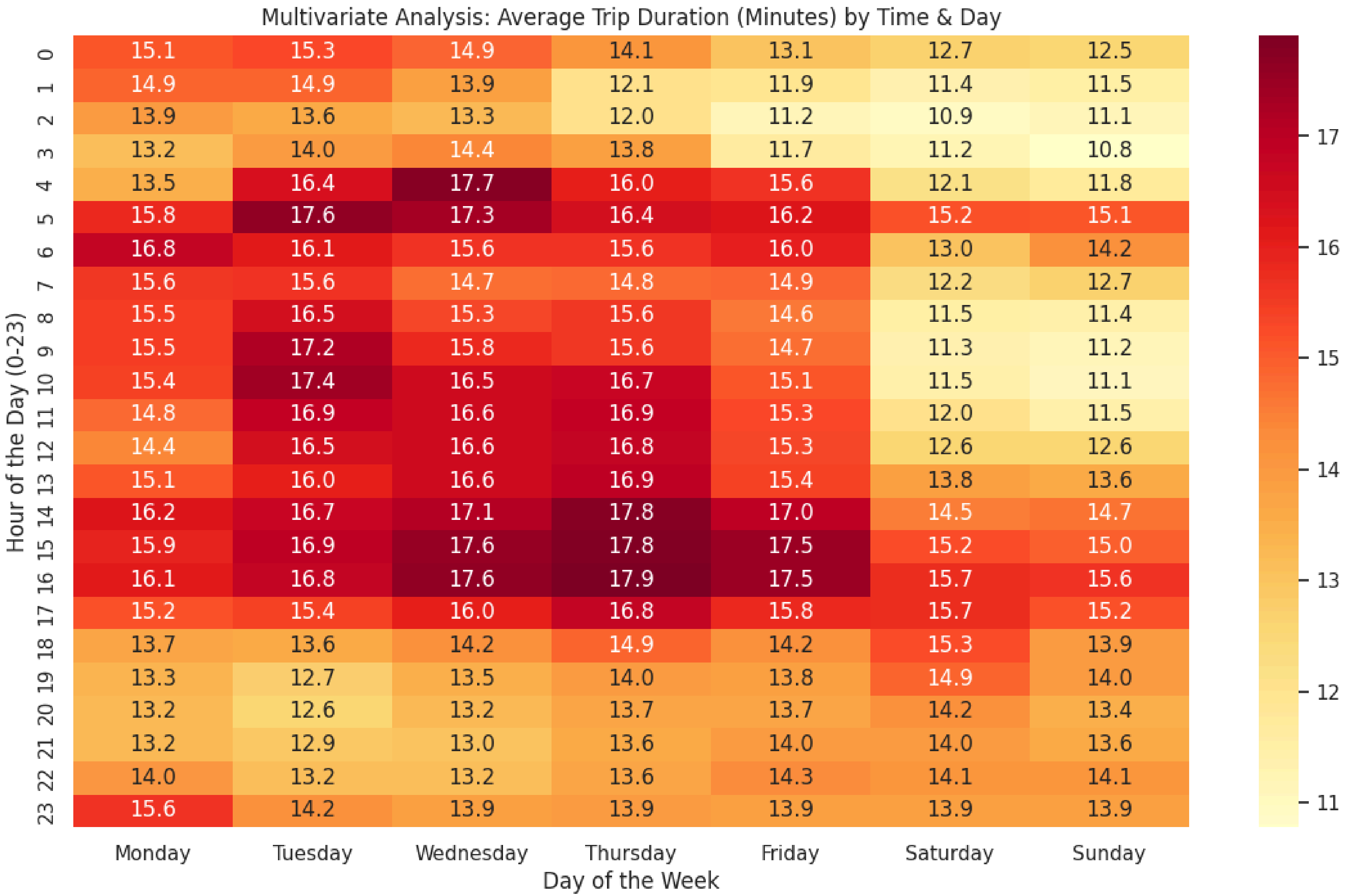
Hexbin analysis reveals that while most urban trips (0–5 miles) conclude within 25 minutes, significant temporal volatility exists. Specifically, 2–3 mile trips spanning 5–150 minutes indicate severe idling and systemic congestion. This erratic duration-to-distance ratio confirms the inefficiency of current traffic management, necessitating the implementation of an adaptive signal control system to optimize flow.



While most trips follow a linear duration-to-fare trajectory, two key anomalies emerge. First, high-fare clusters ($20+) for short distances (2–15 miles) highlight severe congestion-induced idling. Second, the horizontal density at $70 points to a fixed-rate policy for long-duration trips, such as JFK airport transfers, regardless of traffic conditions or time of day.



A comparison of midnight (free-flow) and 5:00 PM (stress) conditions highlights significant system inefficiency. During rush hour, 5-mile transit times increase to 40-80 minutes, a sharp contrast to the baseline, confirming that the current static management system cannot handle peak loads.



Multivariate Analysis: Average Trip Duration (Minutes) by Time & Day
Hour of the Day (0-23)
Day of the Week
Monday Tuesday Wednesday Thursday Friday Saturday Sunday
0 15.1 15.3 14.9 14.1 13.1 12.7 12.5
1 14.9 14.9 13.9 12.1 11.9 11.4 11.5
2 13.9 13.6 13.3 12.0 11.2 10.9 11.1
3 13.2 14.0 14.4 13.8 11.7 11.2 10.8
4 13.5 16.4 17.7 16.0 15.6 12.1 11.8
5 15.8 17.6 17.3 16.4 16.2 15.2 15.1
6 16.8 16.1 15.6 15.6 16.0 13.0 14.2
7 15.6 15.6 14.7 14.8 14.9 12.2 12.7
8 15.5 16.5 15.3 15.6 14.6 11.5 11.4
9 15.5 17.2 15.8 15.6 14.7 11.3 11.2
10 15.4 17.4 16.5 16.7 15.1 11.5 11.1
11 14.8 16.9 16.6 16.9 15.3 12.0 11.5
12 14.4 16.5 16.6 16.8 15.3 12.6 12.6
13 15.1 16.0 16.6 16.9 15.4 13.8 13.6
14 16.2 16.7 17.1 17.8 17.0 14.5 14.7
15 15.9 16.9 17.6 17.8 17.5 15.2 15.0
16 16.1 16.8 17.6 17.9 17.5 15.7 15.6
17 15.2 15.4 16.0 16.8 15.8 15.7 15.2
18 13.7 13.6 14.2 14.9 14.2 15.3 13.9
19 13.3 12.7 13.5 14.0 13.8 14.9 14.0
20 13.2 12.6 13.2 13.7 13.7 14.2 13.4
21 13.2 12.9 13.0 13.6 14.0 14.0 13.6
22 14.0 13.2 13.2 13.6 14.3 14.1 14.1
23 15.6 14.2 13.9 13.9 13.9 13.9 13.9
17
16
15
14
13
12
11

The heatmap confirms that traffic strain peaks midweek (Tuesday–Thursday) from 4:00 PM to 6:00 PM. Mondays show mild congestion, while weekends remain largely clear.